

Py

Day 06/55 - Numerical Operations in IDLE

Use Python as a calculator with REPL, operators, and memory

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Video Summary

This video introduces numerical operations in Python using IDLE, the Integrated Development and Learning Environment. It explains why IDLE is useful for instant code snippets, then moves through arithmetic, order of operations, division styles, variables, and the underscore `_` shortcut for the last computed value.

1 Environment Setup

The lesson separates Command Prompt from Python IDLE. Command Prompt runs system-level commands, while Python IDLE gives a REPL where Python expressions can be tested immediately.



Command Prompt

Useful for system commands such as launching Python, checking versions, and navigating folders.



Python IDLE

A beginner-friendly Python shell for quick expressions, calculations, and instant feedback.

2 REPL: Read-Eval-Print Loop

IDLE works like a calculator for Python expressions: type at the prompt, press Enter, and Python evaluates the expression and prints the result.

1

Read

Take input from the prompt.

2

Eval

Evaluate the expression.

3

Print

Show the result instantly.

4

Loop

Wait for the next input.

3 Arithmetic Basics

Python supports familiar arithmetic operators. In IDLE, each expression can be typed directly after the `>>>` prompt and evaluated instantly.

Operator	Meaning	Example
+	Addition	<code>2 + 3 -> 5</code>
-	Subtraction	<code>9 - 4 -> 5</code>
*	Multiplication	<code>6 * 7 -> 42</code>
/	Standard division	<code>10 / 4 -> 2.5</code>

4 Order of Operations

Python follows mathematical precedence rules similar to BODMAS. Parentheses `()` are the clearest way to control which operation runs first.



```
Without parentheses

>>> 2 + 3 * 4
14
```

```
With parentheses and powers

>>> (2 + 3) * 4
20
>>> 2 ** 3
8
```

5 Division Types

Python gives separate operators for normal division, floor division, and remainders. This is useful when you need either the decimal result, the quotient, or the leftover value.

Operator	Purpose	Example
/	Standard division returns a float.	10 / 3 -> 3.333...
//	Floor division returns the quotient.	10 // 3 -> 3
%	Modulus returns the remainder.	10 % 3 -> 1

6 Variables and Memory

Variables let you store results and reuse them in later calculations. IDLE also keeps the last computed value available through the underscore (`_`) symbol.

Variables

```
>>> x = 5
>>> y = 3
>>> total = x + y
>>> total
8
```

Last value with `_`

```
>>> 10 + 5
15
>>> _ * 2
30
```

Day 06/55 takeaway

IDLE is a friendly place to explore Python as a calculator. Learn the operators, control precedence with parentheses, store values in variables, and use `_` to reuse your last result.